Lctr 13: Fourier Transform Properties, Convolution, and Systems

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Focus: Sections 4.3–4.7 and 5.3–5.8 (Pages 300–340, 372–416)

13.1 The Big Picture: Why Properties Matter

Why This Matters

In Lecture 12 we derived the Fourier transform and computed several basic pairs by evaluating integrals directly. That approach works but becomes tedious for complex signals. The **properties** of the Fourier transform let us build new transform pairs from known ones—without recomputing integrals from scratch. The most important property is the **convolution property**: it turns convolution in time into simple multiplication in frequency. This is the payoff of frequency-domain analysis—it replaces a difficult integral (convolution) with an easy operation (multiplication). Combined with the ability to solve differential and difference equations algebraically, these tools make the Fourier transform the workhorse of signals and systems analysis.

Roadmap for this lecture:

1. Properties of the CT Fourier Transform (with DT parallels noted throughout).
2. The Convolution Property (CT and DT)—the most important property.
3. The Multiplication Property (CT and DT)—the dual of convolution.
4. DT-specific: duality between the DTFT and the DT Fourier series.
5. Systems characterized by differential equations (CT) and difference equations (DT).

13.2 Properties of the CT Fourier Transform

Throughout this section, we use the shorthand $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$.

13.2.1 Linearity

$$\boxed{ax(t) + by(t) \overset{\mathcal{F}}{\longleftrightarrow} aX(j\omega) + bY(j\omega)}$$

Meaning: The FT of a linear combination is the same linear combination of the individual FTs. This is the same linearity property we had for the Fourier series (Lecture 10), now extended to the transform.

13.2.2 Time Shifting

$$\boxed{x(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} e^{-j\omega t_0} X(j\omega)}$$

Meaning: A time delay of t_0 multiplies the spectrum by a linear-phase factor $e^{-j\omega t_0}$. The **magnitude spectrum does not change**—only the phase changes. This generalizes the FS time-shifting property from Lecture 10.

Time Shifting: Delayed Exponential Pulse

From Lecture 12: $e^{-2t}u(t) \overset{\mathcal{F}}{\longleftrightarrow} 1/(2+j\omega)$.
Problem: Find the FT of $x(t) = e^{-2(t-3)}u(t-3)$ (the same pulse, delayed by 3 seconds).
Solution: This is $x(t) = g(t-3)$ where $g(t) = e^{-2t}u(t)$. By time shifting:

$$X(j\omega) = e^{-j3\omega} \cdot \frac{1}{2+j\omega}$$

Check: $|X(j\omega)| = |e^{-j3\omega}| \cdot \frac{1}{\sqrt{4+\omega^2}} = \frac{1}{\sqrt{4+\omega^2}}$. The magnitude is *unchanged*—only the phase acquires an extra -3ω term.

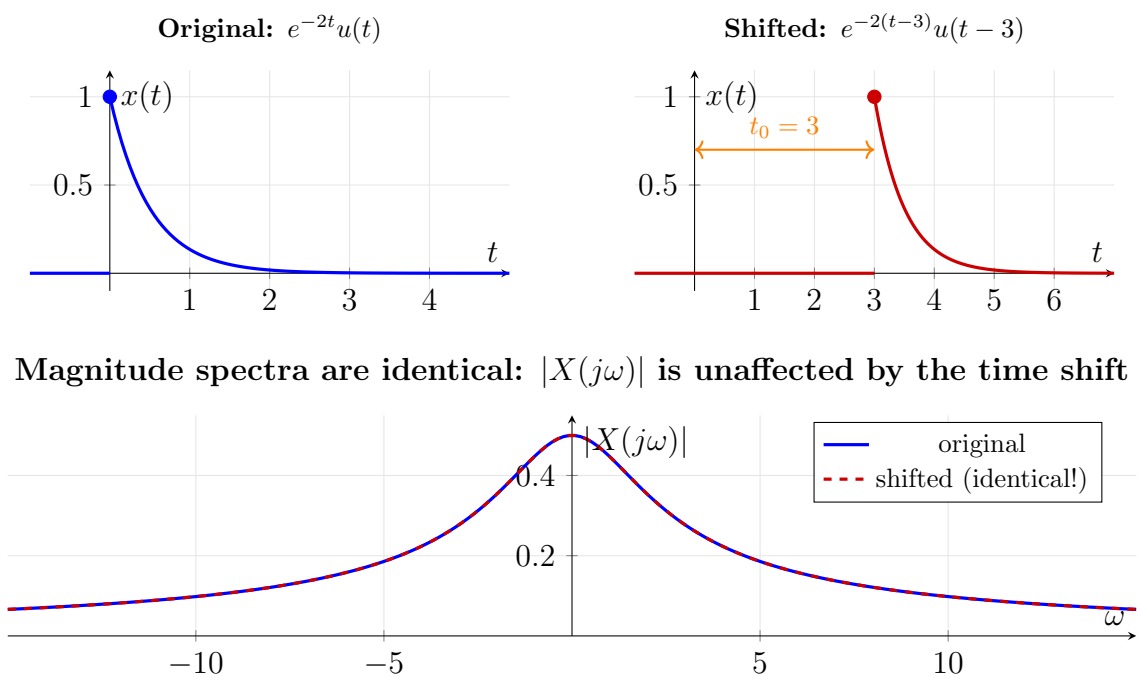

Figure 1: Time shifting does not change the magnitude spectrum. The two curves overlap perfectly.

13.2.3 Frequency Shifting (Modulation)

$$\boxed{e^{j\omega_0 t}\, x(t) \overset{\mathcal{F}}{\longleftrightarrow} X\big(j(\omega-\omega_0)\big)}$$

Meaning: Multiplying a signal by a complex exponential in time shifts its spectrum in frequency. This is the basis of **modulation** in communications—it moves a signal's spectrum to a new center frequency.

Frequency Shifting in Action

Let $x(t) = e^{-|t|}$, so $X(j\omega) = 2/(1+\omega^2)$ (from Lecture 12, with $a = 1$).
Problem: Find the FT of $y(t) = e^{-|t|}\cos(5t)$.
Step 1: Write cosine using Euler: $\cos(5t) = \frac{1}{2}e^{j5t} + \frac{1}{2}e^{-j5t}$.
Step 2: Apply frequency shifting and linearity:

$$Y(j\omega) = \frac{1}{2}X\big(j(\omega-5)\big) + \frac{1}{2}X\big(j(\omega+5)\big) = \frac{1}{1+(\omega-5)^2} + \frac{1}{1+(\omega+5)^2}$$

The original spectrum (centered at $\omega = 0$) has been split into two copies centered at $\omega = \pm 5$.

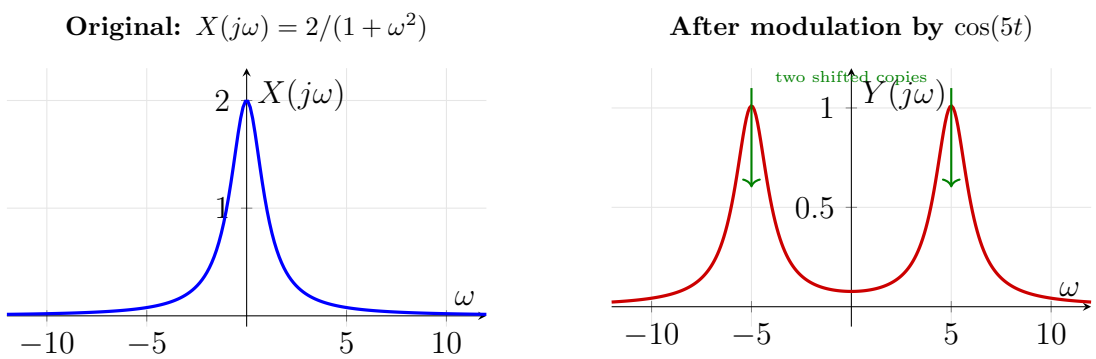

Figure 2: Frequency shifting: multiplying by $\cos(5t)$ splits the spectrum into two copies at $\omega = \pm 5$. This is how AM radio works—the baseband signal is shifted to a carrier frequency.

13.2.4 Conjugation and Conjugate Symmetry

$$x^*(t) \overset{\mathcal{F}}{\longleftrightarrow} X^*(-j\omega)$$

For **real** signals ($x(t) = x^*(t)$), this gives **conjugate symmetry**:

$$\boxed{X(-j\omega) = X^*(j\omega) \quad \text{(when } x(t) \text{ is real)}}$$

Consequences for real signals:

- $|X(j\omega)| = |X(-j\omega)|$ (magnitude is **even**)
- $\angle X(j\omega) = -\angle X(-j\omega)$ (phase is **odd**)
- $\mathrm{Re}\{X(j\omega)\}$ is even; $\mathrm{Im}\{X(j\omega)\}$ is odd

Key Insight

Conjugate symmetry means that for real signals, the negative-frequency content is completely determined by the positive-frequency content. This is why engineers often plot spectra for $\omega \geq 0$ only—the other half carries no new information.

13.2.5 Differentiation and Integration

$$\boxed{\frac{dx(t)}{dt} \overset{\mathcal{F}}{\longleftrightarrow} j\omega\, X(j\omega)}$$

Meaning: Differentiation in time corresponds to multiplication by $j\omega$ in frequency. Higher frequencies are amplified (scaled by $|\omega|$), which is why differentiation enhances rapid changes and edges.

More generally, nth-order differentiation gives:

$$\frac{d^n x(t)}{dt^n} \overset{\mathcal{F}}{\longleftrightarrow} (j\omega)^n\, X(j\omega)$$

For integration (under appropriate convergence conditions):

$$\boxed{\int_{-\infty}^{t} x(\tau)\, d\tau \overset{\mathcal{F}}{\longleftrightarrow} \frac{X(j\omega)}{j\omega} + \pi\, X(0)\, \delta(\omega)}$$

Differentiation Property: From Triangular to Rectangular

The **triangular pulse** $\Lambda(t)$ (height 1, base from -1 to 1) has a known FT: $X(j\omega) = \mathrm{sinc}^2(\omega/2)$ (i.e., the square of a sinc).
Its derivative is a pair of rectangular pulses:

$$\frac{d\Lambda}{dt} = \begin{cases} +1, & -1 < t < 0 \\ -1, & 0 < t < 1 \end{cases}$$

By the differentiation property, the FT of $d\Lambda/dt$ is $j\omega \cdot \mathrm{sinc}^2(\omega/2)$, which must also equal the FT computed directly. This provides a quick consistency check or a way to find one transform from the other.

13.2.6 Time and Frequency Scaling

$$\boxed{x(at) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)}$$

Meaning: Compressing a signal in time ($|a| > 1$) *expands* its spectrum and reduces its amplitude, and vice versa. This is another statement of the **time–frequency trade-off**.

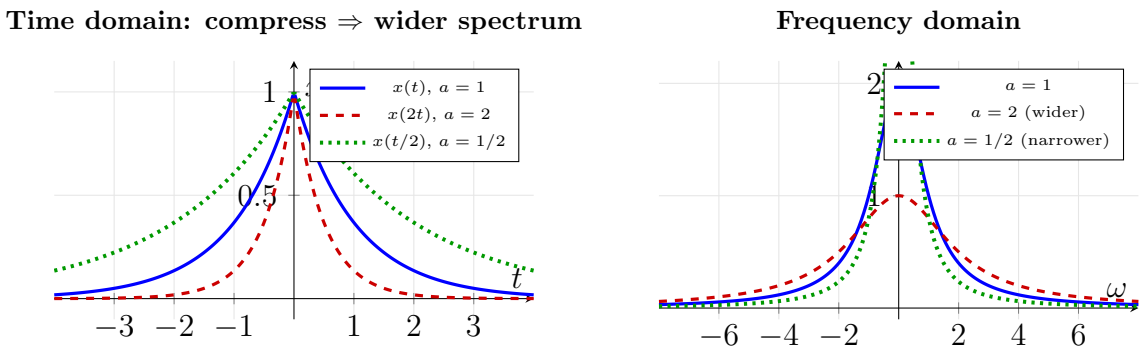

Figure 3: Time–frequency scaling: compressing in time (red, $a = 2$) widens the spectrum but reduces its peak. Stretching in time (green, $a = 1/2$) narrows the spectrum and increases its peak. The area under the magnitude spectrum is conserved (Parseval).

13.2.7 Duality

The CT Fourier transform has a beautiful symmetry. If $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(j\omega)$, then:

$$\boxed{X(jt) \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\, x(-\omega)}$$

Meaning: If you know a transform pair, you automatically get a *second* pair for free by swapping the roles of time and frequency.

Duality: From Rectangular Pulse to Sinc Spectrum

We know: rectangular pulse in time $\overset{\mathcal{F}}{\longleftrightarrow}$ sinc in frequency.
By duality: **sinc in time** $\overset{\mathcal{F}}{\longleftrightarrow}$ **rectangular pulse in frequency**.
Specifically, if $x(t) = \frac{\sin(Wt)}{\pi t}$ (a sinc function), then:

$$X(j\omega) = \begin{cases} 1, & |\omega| \le W \\ 0, & |\omega| > W \end{cases}$$

This is an **ideal lowpass filter** with cutoff frequency W! The sinc function is its impulse response—something we will use extensively in Lectures 14–15.

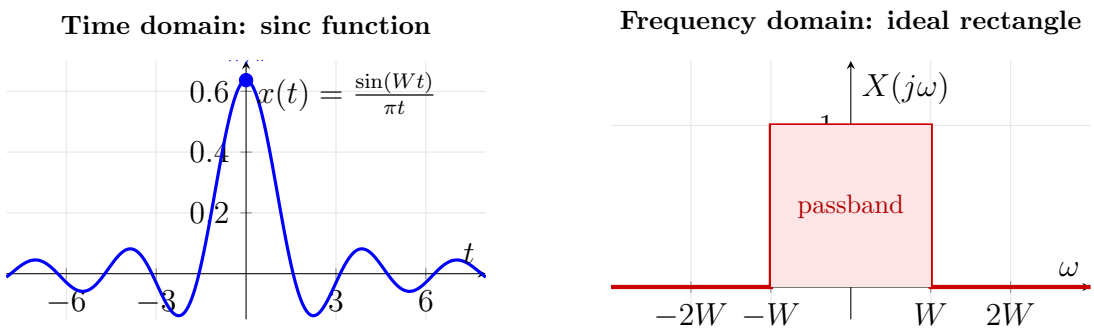

Figure 4: Duality in action: a sinc in time transforms into a rectangular spectrum—the impulse response of an ideal lowpass filter.

13.2.8 Parseval's Relation (Energy Conservation)

$$\boxed{\int_{-\infty}^{+\infty} |x(t)|^2 \, dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} |X(j\omega)|^2 \, d\omega}$$

Meaning: The total energy of the signal can be computed either in the time domain (left side) or the frequency domain (right side). The quantity $|X(j\omega)|^2$ is the **energy spectral density**—it tells us how the signal's energy is distributed across frequencies.

This generalizes the Parseval relation for the Fourier series (Lecture 10), which dealt with average *power* of periodic signals. Here we deal with total *energy* of aperiodic signals.

13.3 Properties of the DT Fourier Transform (DTFT)

The DTFT properties parallel those of the CT FT, with minor notational differences. We summarize them in a side-by-side table, then discuss the important DT-specific properties.

Property	**CT Fourier Transform**	**DTFT**
Linearity	$ax + by \leftrightarrow aX + bY$	$ax + by \leftrightarrow aX + bY$
Time shift	$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X$	$x[n - n_0] \leftrightarrow e^{-j\omega n_0} X$
Freq. shift	$e^{j\omega_0 t} x \leftrightarrow X(j(\omega - \omega_0))$	$e^{j\omega_0 n} x \leftrightarrow X(e^{j(\omega - \omega_0)})$
Conjugate sym. (real x)	$X(-j\omega) = X^*(j\omega)$	$X(e^{-j\omega}) = X^*(e^{j\omega})$
Differentiation / Differencing	$\frac{dx}{dt} \leftrightarrow j\omega X$	$x[n] - x[n-1] \leftrightarrow (1 - e^{-j\omega})X$
Time scaling	$x(at) \leftrightarrow \frac{1}{\lvert a \rvert} X(j\omega/a)$	Time expansion (see below)
Parseval	$\int \lvert x \rvert^2 \, dt = \frac{1}{2\pi} \int \lvert X \rvert^2 \, d\omega$	$\sum \lvert x[n] \rvert^2 = \frac{1}{2\pi} \int_{2\pi} \lvert X \rvert^2 \, d\omega$

13.3.1 DT Time Expansion

In discrete time, we cannot simply "compress" a signal by a non-integer factor. Instead, the DT time-expansion property inserts zeros:

If $x_{(k)}[n] = \begin{cases} x[n/k], & n = 0, \pm k, \pm 2k, \ldots \\ 0, & \text{otherwise} \end{cases}$

then $x_{(k)}[n] \overset{\mathcal{F}}{\longleftrightarrow} X(e^{jk\omega})$.

Meaning: Inserting $k-1$ zeros between each sample *compresses* the spectrum by a factor of k (making it repeat k times within $[0, 2\pi)$). This is important for multirate signal processing and sampling rate conversion.

13.3.2 Frequency Differentiation (DT-specific)

$$\boxed{n\,x[n] \overset{\mathcal{F}}{\longleftrightarrow} j\frac{dX(e^{j\omega})}{d\omega}}$$

Multiplying by n in the time domain corresponds to differentiating in frequency (times j). There is no simple CT analog at the same level of usefulness.

13.4 The Convolution Property

Why This Matters

This is arguably the **most important property** in this entire course. It states that convolution in the time domain becomes multiplication in the frequency domain. This means that instead of computing a difficult convolution integral (or sum), we can: (1) transform to frequency, (2) multiply, (3) transform back. More fundamentally, it gives us $Y(j\omega) = X(j\omega) \cdot H(j\omega)$ for *any* input to an LTI system—not just periodic inputs (as we had in Lecture 11).

13.4.1 CT Convolution Property

$$\boxed{y(t) = x(t) * h(t) \overset{\mathcal{F}}{\longleftrightarrow} Y(j\omega) = X(j\omega) \cdot H(j\omega)}$$

Proof sketch: Start from $Y(j\omega) = \int y(t)\, e^{-j\omega t}\, dt = \int \left[\int x(\tau)\, h(t-\tau)\, d\tau\right] e^{-j\omega t}\, dt$. Switch the order of integration, substitute $t' = t - \tau$, and the double integral separates into $X(j\omega) \cdot H(j\omega)$.

Visual Intuition

The convolution property says: the **spectrum of the output** is the **product** of the input spectrum and the frequency response. At each frequency ω, the system independently scales the input by $|H(j\omega)|$ and shifts its phase by $\angle H(j\omega)$.

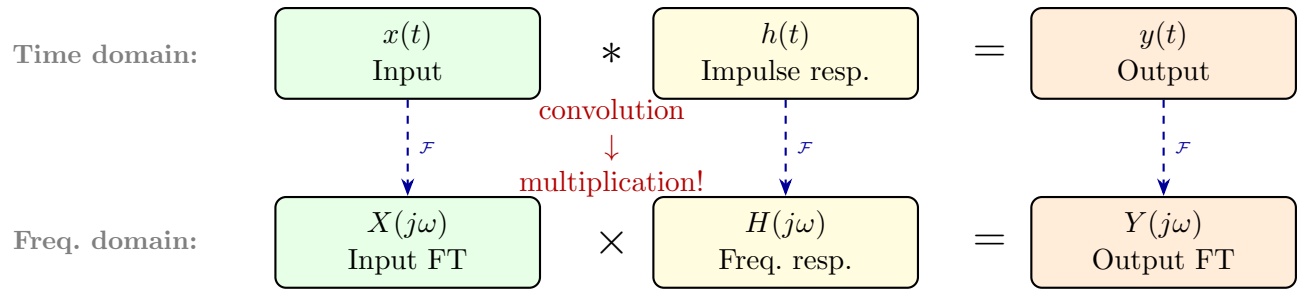

Figure 5: The convolution property: the difficult operation (convolution, $*$) in the time domain becomes the easy operation (multiplication, $\times$) in the frequency domain. This is why we use the Fourier transform.

Convolution Property: Finding System Output

System: $h(t) = e^{-3t}u(t)$, so $H(j\omega) = \frac{1}{3+j\omega}$.
Input: $x(t) = e^{-2t}u(t)$, so $X(j\omega) = \frac{1}{2+j\omega}$.
Step 1: Output spectrum: $Y(j\omega) = X(j\omega) \cdot H(j\omega) = \frac{1}{(2+j\omega)(3+j\omega)}$.
Step 2: Partial fraction expansion:

$$Y(j\omega) = \frac{1}{(2+j\omega)(3+j\omega)} = \frac{A}{2+j\omega} + \frac{B}{3+j\omega}$$

Solve: $A(3+j\omega) + B(2+j\omega) = 1$. Set $j\omega = -2$: $A = 1$. Set $j\omega = -3$: $B = -1$.

$$Y(j\omega) = \frac{1}{2+j\omega} - \frac{1}{3+j\omega}$$

Step 3: Inverse FT (recognize each term from Lecture 12):

$$y(t) = e^{-2t}u(t) - e^{-3t}u(t) = \left(e^{-2t} - e^{-3t}\right)u(t)$$

Check: At $t = 0$: $y(0) = 1 - 1 = 0$ ✓ (the output starts at zero, as expected for two causal signals convolved).
At $t = 1$: $y(1) = e^{-2} - e^{-3} = 0.135 - 0.050 = 0.085$.

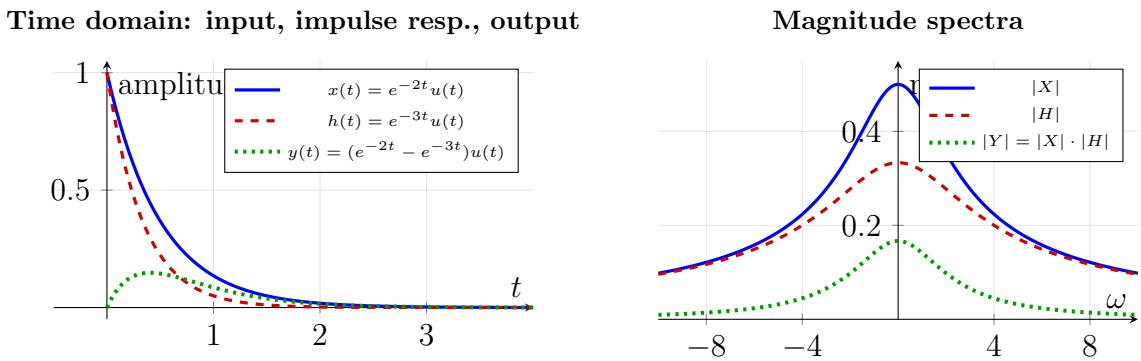

Figure 6: Convolution in time $\leftrightarrow$ multiplication in frequency. The output magnitude $|Y|$ is the *product* of $|X|$ and $|H|$ at each frequency.

13.4.2 DT Convolution Property

The exact same structure holds in discrete time:

$$\boxed{y[n] = x[n] * h[n] \overset{\mathcal{F}}{\longleftrightarrow} Y(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega})}$$

DT Convolution Property Example

System: $h[n] = (0.5)^n u[n]$, so $H(e^{j\omega}) = \frac{1}{1-0.5\,e^{-j\omega}}$.
Input: $x[n] = (0.8)^n u[n]$, so $X(e^{j\omega}) = \frac{1}{1-0.8\,e^{-j\omega}}$.
Step 1: Output DTFT:

$$Y(e^{j\omega}) = \frac{1}{(1-0.5\,e^{-j\omega})(1-0.8\,e^{-j\omega})}$$

Step 2: Partial fractions (in the variable $e^{-j\omega}$, or equivalently in z^{-1}):

$$Y(e^{j\omega}) = \frac{A}{1-0.5\,e^{-j\omega}} + \frac{B}{1-0.8\,e^{-j\omega}}$$

Setting $e^{-j\omega} = 1/0.5 = 2$: $A(1-0.8\cdot 2) = 1 \Rightarrow A(-0.6) = 1 \Rightarrow A = -5/3$.
Setting $e^{-j\omega} = 1/0.8 = 1.25$: $B(1-0.5\cdot 1.25) = 1 \Rightarrow B(0.375) = 1 \Rightarrow B = 8/3$.
Step 3: Inverse DTFT:

$$y[n] = -\frac{5}{3}(0.5)^n u[n] + \frac{8}{3}(0.8)^n u[n]$$

Check: $y[0] = -5/3 + 8/3 = 3/3 = 1$ ✓ (since $x[0]\cdot h[0] = 1\cdot 1 = 1$ is the first term of the convolution).

13.5 The Multiplication Property

The multiplication property is the **dual** of the convolution property: multiplication in time corresponds to convolution in frequency.

13.5.1 CT Multiplication Property

$$\boxed{r(t) = x(t) \cdot y(t) \overset{\mathcal{F}}{\longleftrightarrow} R(j\omega) = \frac{1}{2\pi} X(j\omega) * Y(j\omega) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\theta)\, Y\big(j(\omega - \theta)\big)\, d\theta}$$

Meaning: Multiplying two signals in time means *convolving* their spectra in frequency (with a $1/(2\pi)$ scaling factor).

Key Insight

The multiplication property is the mathematical foundation of:

- **Amplitude modulation (AM):** Multiplying a signal by $\cos(\omega_c t)$ shifts its spectrum to $\pm\omega_c$ (frequency shifting from Section 13.2.3 is a special case).
- **Sampling:** Multiplying a signal by a periodic impulse train (Chapter 7).
- **Windowing:** Multiplying a signal by a window function "smears" its spectrum (due to convolution with the window's FT).

13.5.2 DT Multiplication Property

$$\boxed{r[n] = x[n] \cdot y[n] \overset{\mathcal{F}}{\longleftrightarrow} R(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta})\, Y\big(e^{j(\omega-\theta)}\big)\, d\theta}$$

The same idea: multiplication in time = periodic convolution in frequency. The integration is over one period of 2π (since DTFT is periodic).

13.6 Duality Between the DTFT and the DT Fourier Series

Why This Matters

In continuous time, duality is between the FT and itself (both involve integrals over continuous variables). In discrete time, duality connects *two different* representations: the **DTFT** and the **DT Fourier Series**. This is because the DTFT spectrum $X(e^{j\omega})$ is periodic in ω (like a periodic signal is in time), so it can itself be represented as a Fourier series.

The key relationship: the DTFT analysis equation

$$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n]\, e^{-j\omega n}$$

looks exactly like a *Fourier series* of $X(e^{j\omega})$ viewed as a periodic function of ω with period 2π. In this interpretation, $x[n]$ plays the role of Fourier coefficients, and n plays the role of the harmonic index.

Conversely, if we have a periodic signal $\tilde{x}[n]$ with period N and FS coefficients a_k, the DT FS synthesis equation

$$\tilde{x}[n] = \sum_{k=0}^{N-1} a_k\, e^{jk(2\pi/N)n}$$

looks like an "inverse DTFT" with a_k playing the role of a finite-length sequence.

Key Insight

DT Duality Summary:

- An aperiodic DT signal $x[n]$ has a **periodic, continuous** spectrum $X(e^{j\omega})$ (DTFT).
- A periodic DT signal $\tilde{x}[n]$ has a **periodic, discrete** spectrum a_k (FS).

There is no "continuous-continuous" duality as in CT—discrete time always produces periodicity somewhere.

13.7 Systems from Differential and Difference Equations

Why This Matters

In Lecture 11 we found the frequency response of a specific RC circuit by substituting $e^{j\omega t}$ into its differential equation. Now we generalize: the Fourier transform turns *any* linear constant-coefficient differential equation (CT) or difference equation (DT) into an *algebraic* equation that we can solve for $H(j\omega)$ or $H(e^{j\omega})$ directly. No more guessing solutions or solving characteristic equations—just algebra.

13.7.1 CT Systems: Differential Equations

Consider a general CT LTI system described by:

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

Step 1: Take the Fourier transform of both sides. Using the differentiation property ($d^k/dt^k \leftrightarrow (j\omega)^k$):

$$\left[\sum_{k=0}^{N} a_k \left(j\omega\right)^k\right] Y(j\omega) = \left[\sum_{k=0}^{M} b_k \left(j\omega\right)^k\right] X(j\omega)$$

Step 2: Solve for $H(j\omega) = Y(j\omega)/X(j\omega)$:

$$\boxed{H(j\omega) = \frac{\sum_{k=0}^{M} b_k \left(j\omega\right)^k}{\sum_{k=0}^{N} a_k \left(j\omega\right)^k}}$$

The rule: Replace every d/dt with $j\omega$, then solve algebraically.

Second-Order CT System from a Differential Equation

System: $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y(t) = x(t)$
Step 1: Replace $d/dt \to j\omega$, $d^2/dt^2 \to (j\omega)^2$:

$$(j\omega)^2 Y + 5(j\omega)Y + 6Y = X$$

Step 2: Factor and solve:

$$H(j\omega) = \frac{1}{(j\omega)^2 + 5(j\omega) + 6} = \frac{1}{-\omega^2 + j5\omega + 6}$$

Step 3: Factor the denominator: $-\omega^2 + j5\omega + 6 = (j\omega + 2)(j\omega + 3)$.

$$H(j\omega) = \frac{1}{(j\omega + 2)(j\omega + 3)}$$

Step 4: Partial fractions (exactly as in the convolution example!):

$$H(j\omega) = \frac{1}{j\omega + 2} - \frac{1}{j\omega + 3}$$

Step 5: Inverse FT gives the impulse response:

$$h(t) = \left(e^{-2t} - e^{-3t}\right) u(t)$$

Check: The system is **stable** because both exponentials decay ($a = 2, 3 > 0$). The frequency response $|H(j\omega)|$ peaks near $\omega = 0$ (lowpass behavior).

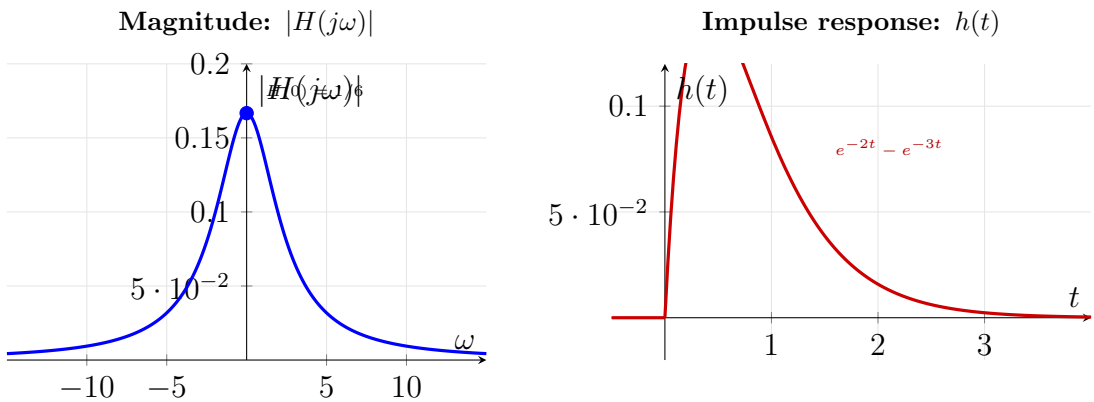

Figure 7: Second-order system: the frequency response shows lowpass behavior (DC gain $= 1/6$, rolls off for large $|\omega|$). The impulse response rises from zero, peaks near $t \approx 0.4$, then decays.

13.7.2 DT Systems: Difference Equations

Consider a general DT LTI system:

$$\sum_{k=0}^{N} a_k \, y[n-k] = \sum_{k=0}^{M} b_k \, x[n-k]$$

Step 1: Take the DTFT. Using the time-shifting property ($x[n-k] \leftrightarrow e^{-jk\omega} X$):

$$\left[\sum_{k=0}^{N} a_k \, e^{-jk\omega}\right] Y(e^{j\omega}) = \left[\sum_{k=0}^{M} b_k \, e^{-jk\omega}\right] X(e^{j\omega})$$

Step 2: Solve for $H(e^{j\omega})$:

$$\boxed{H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k \, e^{-jk\omega}}{\sum_{k=0}^{N} a_k \, e^{-jk\omega}}}$$

The rule: Replace every delay z^{-1} (i.e., $n \to n-1$) with $e^{-j\omega}$, then solve algebraically.

Second-Order DT System

System: $y[n] - 1.2y[n-1] + 0.32y[n-2] = x[n]$
Step 1: Replace delays with $e^{-j\omega}$:

$$Y(1 - 1.2\,e^{-j\omega} + 0.32\,e^{-j2\omega}) = X$$

Step 2:

$$H(e^{j\omega}) = \frac{1}{1 - 1.2\,e^{-j\omega} + 0.32\,e^{-j2\omega}}$$

Step 3: Factor the denominator. We need the roots of $1 - 1.2z^{-1} + 0.32z^{-2} = 0$, or equivalently $z^2 - 1.2z + 0.32 = 0$. Using the quadratic formula:

$$z = \frac{1.2 \pm \sqrt{1.44 - 1.28}}{2} = \frac{1.2 \pm 0.4}{2} \implies z_1 = 0.8,\ z_2 = 0.4$$

So: $H(e^{j\omega}) = \frac{1}{(1-0.8\,e^{-j\omega})(1-0.4\,e^{-j\omega})}$
Step 4: Partial fractions:

$$H(e^{j\omega}) = \frac{A}{1 - 0.8\,e^{-j\omega}} + \frac{B}{1 - 0.4\,e^{-j\omega}}$$

Setting $e^{-j\omega} = 1/0.8$: $A(1 - 0.4/0.8) = 1 \Rightarrow A(0.5) = 1 \Rightarrow A = 2$.
Setting $e^{-j\omega} = 1/0.4$: $B(1 - 0.8/0.4) = 1 \Rightarrow B(-1) = 1 \Rightarrow B = -1$.
Step 5: Inverse DTFT:

$$h[n] = 2(0.8)^n u[n] - (0.4)^n u[n]$$

Check: Both poles (0.8 and 0.4) have $|z| < 1$, so the system is **stable**. $h[0] = 2-1 = 1$ ✓ (consistent with $a_0 = 1$ in the difference equation).

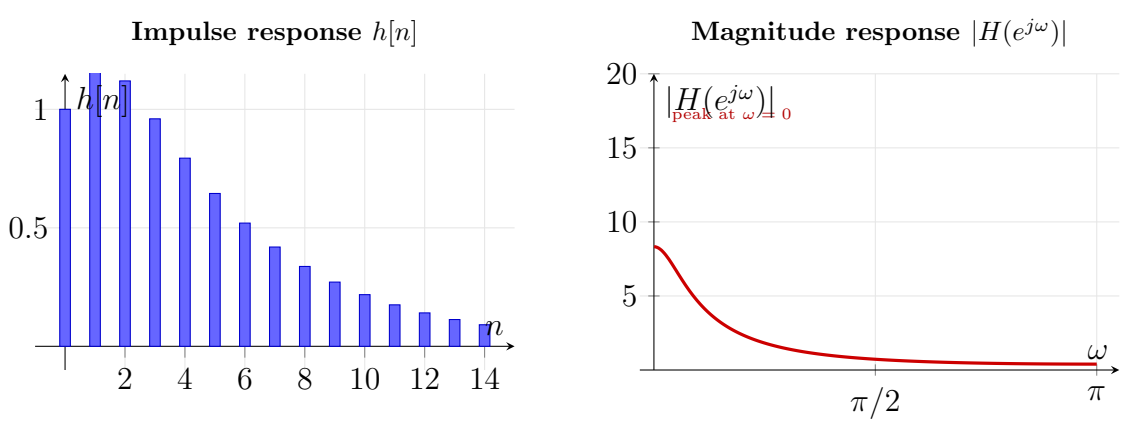

Figure 8: Second-order DT system with poles at $z = 0.8$ and $z = 0.4$. The impulse response decays geometrically. The frequency response shows strong lowpass behavior (large peak at $\omega = 0$, rolls off toward $\omega = \pi$).

13.8 Summary of the Analysis Pipeline

Visual Intuition

With the convolution property and the ability to solve differential/difference equations algebraically, we now have a complete **frequency-domain analysis pipeline**. Given any LTI system and any input, we can find the output using the Fourier transform.

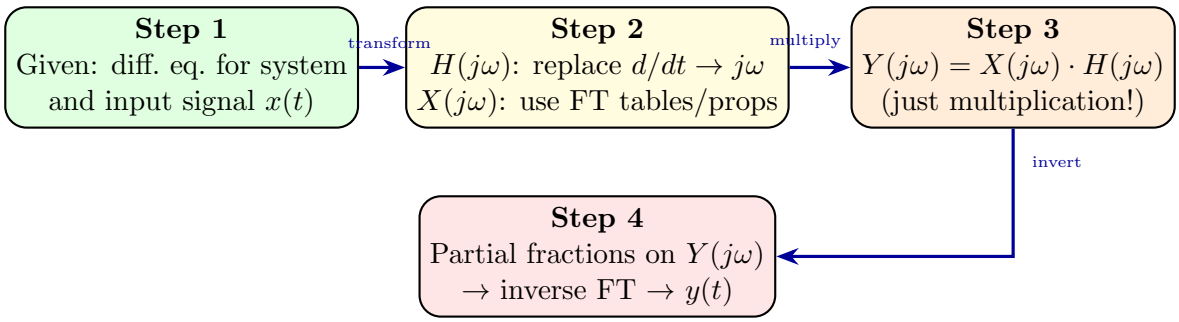

Figure 9: The four-step frequency-domain analysis pipeline. This works for both CT (replace d/dt with $j\omega$) and DT (replace delays with $e^{-j\omega}$).

13.9 Complete Property Reference Tables

13.9.1 CT Fourier Transform Properties

Property	**Signal**	**Fourier Transform**
Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
Time shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
Frequency shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
Conjugation	$x^*(t)$	$X^*(-j\omega)$
Time reversal	$x(-t)$	$X(-j\omega)$
Time scaling	$x(at)$	$\frac{1}{\lvert a \rvert} X(j\omega/a)$
Differentiation (time)	$\frac{dx}{dt}$	$j\omega\, X(j\omega)$
Differentiation (freq.)	$-jt\, x(t)$	$\frac{dX(j\omega)}{d\omega}$
Convolution	$x(t) * h(t)$	$X(j\omega) \cdot H(j\omega)$
Multiplication	$x(t) \cdot y(t)$	$\frac{1}{2\pi} X(j\omega) * Y(j\omega)$
Duality	$X(jt)$	$2\pi\, x(-\omega)$
Parseval	$\int \lvert x \rvert^2\, dt$	$= \frac{1}{2\pi} \int \lvert X \rvert^2\, d\omega$

13.9.2 DTFT Properties

Property	**Signal**	**DTFT**
Linearity	$ax[n] + by[n]$	$aX + bY$
Time shifting	$x[n - n_0]$	$e^{-j\omega n_0} X(e^{j\omega})$
Frequency shifting	$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$
Conjugation	$x^*[n]$	$X^*(e^{-j\omega})$
Time reversal	$x[-n]$	$X(e^{-j\omega})$
First difference	$x[n] - x[n-1]$	$(1 - e^{-j\omega})X$
Freq. differentiation	$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$
Convolution	$x[n] * h[n]$	$X(e^{j\omega}) \cdot H(e^{j\omega})$
Multiplication	$x[n] \cdot y[n]$	$\frac{1}{2\pi} \int_{2\pi} X \cdot Y\, d\theta$
Parseval	$\sum \lvert x[n] \rvert^2$	$= \frac{1}{2\pi} \int_{2\pi} \lvert X \rvert^2\, d\omega$

13.10 Common Mistakes to Avoid

1. **Confusing convolution ($*$) with multiplication ($\cdot$):** Convolution in time $\leftrightarrow$ multiplication in frequency. Multiplication in time $\leftrightarrow$ convolution in frequency. These are *opposite* directions. Mixing them up is the single most common error in frequency-domain analysis.

2. **Forgetting the $1/(2\pi)$ in the multiplication property:** When two signals are multiplied in time, their spectra are convolved *and* divided by 2π. The convolution property does NOT have this factor—only the multiplication property does.

3. **Wrong sign in time-shift exponent:** A delay of $t_0 > 0$ gives $e^{-j\omega t_0}$ (negative sign in exponent). An advance of t_0 gives $e^{+j\omega t_0}$ (positive sign). The sign matters!

4. **Applying the differentiation property to discontinuous signals without care:** If $x(t)$ has a jump discontinuity, its derivative includes an impulse at the jump. You must account for this when using $dx/dt \leftrightarrow j\omega X$.

5. **Partial fractions with wrong variable in DT:** When factoring DT frequency responses, work with $e^{-j\omega}$ (or z^{-1}), not with $e^{j\omega}$ or ω directly. The poles of the system are the values where $e^{-j\omega} = 1/a_i$, i.e., the coefficients that appear with $e^{-j\omega}$ in the denominator factors.

6. **Forgetting stability when inverting:** The partial fraction $\rightarrow$ inverse FT approach assumes the system is stable (all poles inside the unit circle for DT, or all poles in the left half-plane for CT). If the system is unstable, the inverse FT must use one-sided transforms (Laplace/z-transform), which we will cover later.

Key Insight

The central message of this lecture: The Fourier transform turns convolution into multiplication and differential/difference equations into algebraic equations. This is why frequency-domain analysis is so powerful—it replaces the hardest operations in time-domain analysis with the easiest ones.

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